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s12. INTRUSION DETECTION

AIM

To understand intrusion detection systems (IDS) and study basic techniques used to evade detection during cybersecurity attacks.

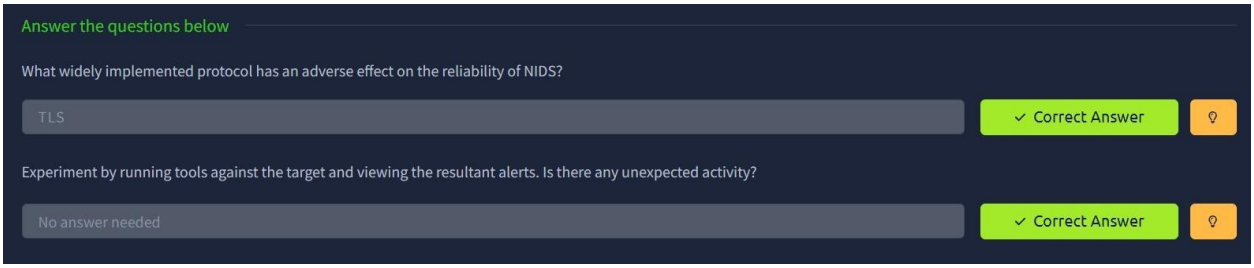
PROCEDURE

1. **Study the concepts and categories of IDS, including Network-based IDS (NIDS) and Host-based IDS (HIDS).**
2. **Perform reconnaissance activities using tools such as Nmap and monitor the generated IDS alerts.**
3. **Explore evasion methods such as stealth scanning and modification of user-agent information.**
4. Compare detection results and examine post-exploitation activities within the monitored environment.

QUESTIONS

TASK 1 - Intrusion Detection Basics

TASK 2 - Network-based IDS (NIDS)



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TASK 3 - Reconnaissance and Evasion Basics

Answer the questions below

What scale is used to measure alert severity in Suricata? (*-*)

1-3 ✓ Correct Answer

How many services is nmap able to fully recognise when the service scan (-sV) is performed?

3 ✓ Correct Answer

TASK 4 - Further Reconnaissance Evasion

Answer the questions below

Nikto, should find an interesting path when the first scan is performed, what is it called?

/login ✓ Correct Answer

What value is used to toggle denial of service vectors when using scan tuning (-T) in nikto?

6 ✓ Correct Answer

Which flags are used to modify the request spacing in nikto? Use commas to separate the flags in your answer.

6,A,B ✓ Correct Answer

TASK 5 - Open Source Intelligence

Answer the questions below

What version of Grafana is the server running?

8.2.5 ✓ Correct Answer

What is the ID of the severe CVE that affects this version of Grafana?

CVE-2021-43798 ✓ Correct Answer

If this server was publicly available, What site might have information on its services already?

shodan ✓ Correct Answer

How would we search the site "example.com" for pdf files, using advanced Google search tags?

site:example.com filetype:pdf ✓ Correct Answer

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TASK 6 - Rulesets

Answer the questions below

What is the password of the grafana-admin account?

GraphingTheWorld32 ✓ Correct Answer

Is it possible to gain direct access to the server now that the grafana-admin password is known? (yay/nay)

yay ✓ Correct Answer

Are any of the attached IDS able to detect the attack if the file /etc/shadow is requested via the exploit, if so what IDS detected it?

Suricata ✓ Correct Answer

TASK 7 - Host Based IDS (HIDS)

Answer the questions below

What category does Wazuh place HTTP 400 error codes in?

web ✓ Correct Answer

Play around with some post-exploitation tools and commands and make note of what activity is detected by Wazuh; compare it to the activity that's detected by Suricata.

No answer needed ✓ Correct Answer

TASK 8 - Privilege Escalation Recon

Answer the questions below

What tool does linPEAS detect as having a potential escalation vector?

docker ✓ Correct Answer

Is an alert triggered by Wazuh when linPEAS is added to the system, if so what its severity?

5 ✓ Correct Answer

TASK 9 - Performing Privilege Escalation

Answer the questions below

What tool does linPEAS detect as having a potential escalation vector?

docker ✓ Correct Answer

Is an alert triggered by Wazuh when linPEAS is added to the system, if so what its severity?

5 ✓ Correct Answer

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RESULT

Understood how IDS detects attacks and how evasion techniques can reduce detection during different attack stages.